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Hierarchical Topic Extraction for Educational Content Representation

Luiza Vidal Copolillo Coelho

Master's in Computer Science and Business Management

Supervisors:
Doctor Ana Rita Peixoto, Assistant Professor,
Iscte – Instituto Universitário de Lisboa

2025/26

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TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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Acknowledgment

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Resumo

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Abstract

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KEYWORDS: *word_1*, *word_2*, ...

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List of Acronyms

SLR: Systematic Literature Review

RQ: Research Question

KW: Keyword

SS: Search String

ML: Machine Learning

TM: Topic Modeling

HTM: Hierarchical Topic Modeling

LDA: Latent Dirichlet Allocation

PLSA: Probabilistic Latent Semantic Analysis

TF-IDF: Term Frequency-Inverse Document Frequency

BoW: Bag of Words

LSA: Latent Semantic Analysis

IoT: Internet of Things

hLDA: hierarchical Latent Dirichlet Allocation

CRP: Chinese Restaurant Process

GHTM: Guided Hierarchical Topic Model

GCTM: Graph Convolutional Topic Model

KR: Knowledge Representation

KG: Knowledge Graph

QA: Question Answering

KGE: Knowledge Graph Embeddings

CRISP-DM: CRoss Industry Standard Process for Data Mining

CHAPTER 1

Introduction

1.1. Contextualization

In the educational context, written tests remain one of the most prominent forms of student assessment. Despite growing interest in alternative evaluation methods, the process of designing, validating, and reviewing traditional assessments remains labor-intensive and often lacks tools for systematic quality assurance [1]–[3]. This challenge is especially evident in the alignment between test questions and course content, which nowadays is commonly conveyed through lecture slides.

To address these issues, the broader project **TestMATE**—Test Making Assistance a Tool for Education—proposes the development of an intelligent support system for test creation. One of its major components involves automatically extracting and representing the knowledge embedded in lecture slides, this task is naturally complex, given the unstructured nature of slide content and the lack of narrative continuity. Within this scope, this research focuses on investigating how Topic Modeling (TM) techniques, especially Hierarchical Topic Modeling (HTM), can be applied to capture and represent educational content effectively. By exploring the integration of TM with Knowledge Representation (KR), we aim to contribute to the automated understanding of curricular content.

1.2. Research Questions and Goals

The central objective of this thesis is to explore the possible benefits of extracting and structuring educational content using advanced TM techniques. In particular, this research seeks to answer the following research question:

- **RQ1:** Is it possible to extract and represent the knowledge of educational content in the format of slides, using topic modeling?

This research aims to evaluate how hierarchical topic structures can enhance the interpretability and organization of slide-based content and to assess the potential of combining such models with KR techniques, such as ontologies or Knowledge Graphs (KGs), to support next-generation educational applications, such as written evaluation test creation.

1.3. Methodology

To answer the Research Question (RQ) and develop a solid foundation for this study, two methodological approaches are employed. Initially, an Systematic Literature Review (SLR) was conducted to survey existing research on: **Topic Modeling**, **Knowledge Representation**, and their **intersection in educational contexts**. The review

followed established guidelines proposed by Kitchenham [4] and Okoli [5] to ensure scientific methodological reliability.

Additionally, for the practical development and evaluation of the proposed solution, we plan to adopt the CRoss Industry Standard Process for Data Mining (CRISP-DM) methodology, proposed by Wirth and Hipp [6]—further detailed in Chapter 3. This methodology is a potential fit for the development of this research because it allows flexibility and iterative refinement as we explore different TM strategies and KR integration approaches.

1.4. Document Structure

This document is structured as follows. Chapter 1 briefly presents the introduction and scope of the research. Chapter 2 reviews the state-of-the-art on **Topic Modeling**, **Knowledge Representation**, and their **application in the educational domains**. Chapter 3 describes the methodological framework adopted.

CHAPTER 2

Literature Review

2.1. Research Methodology

The methodological approach used to conduct the literature review of this work was a SLR. An SLR is a structured approach to identify, evaluate, and synthesize existing research on a specific topic. Unlike traditional research approaches, SLRs specifies precise guidelines and procedures, posteriorly clarified, to reduce bias and guarantee a wide range of relevant studies [7]. In conclusion, this approach improves the scientific reliability and validity of claims and findings, providing a high-quality foundation for evidence-based practice and further research.

In line with this methodology, the review process was structured according to Kitchenham [4] and Okoli [5]. The structure of the review has three phases: planning the review (Subsection 2.1.1), conducting the review (Subsection 2.1.2), and reporting the review (Section 2.2 - 2.6). The three phases are described and illustrated in Table 2.1. These steps ensure a structured and repeatable process, aligned with best practices for systematic reviews as defined in software engineering and information systems research.

Table 2.1. Stages of an SLR [4]

Stage	**Activities**
1. Planning the Review	a. Identification of the need for a review b. Development of a review protocol
2. Conducting the Review	a. Identification of research b. Selection of research c. Study Quality Assessment d. Data Extraction and Monitoring e. Data Synthesis
3. Reporting the Review	a. Review Findings and Conclusions

2.1.1. Planning the Review

Professors write assessments manually, which is time-consuming, often lacking systematic validation and feedback mechanisms [2]. Not only that, but professors occasionally struggle to ensure alignment between test content and curricular goals [3]. More specifically, a key challenge concerns the representation of educational content, particularly that transmitted through lecture slides, within written assessments designed to evaluate student learning. This issue is further aggravated by practical constraints, including limited preparation time and insufficient institutional support.

Additionally, Sireci [1] points out that the absence of structured methods for collecting content validity evidence further weakens the reliability of learning assessment evaluations. For this reason, this review intends to consolidate the existing research on knowledge representation, and educational topic modeling, as a foundation for developing intelligent tools that can support the automatic identification and structuring of curricular content from lecture slides to enhance test design. Therefore, the objective of the present literature review is to provide answers to the following RQ:

RQ: Is it possible to extract and represent the knowledge of educational content in the format of slides using topic modeling?

We created a review protocol to search for answers in the literature. Table 2.2 shows each stage of the protocol. It begins with the definition of keywords based on the research questions defined, which were then combined in different ways to search for publications. The following databases were considered: IEEE Xplore Digital Library[1], ACM Digital Library[2], Scopus[3], and Web of Science[4]. To complement what was presented in 2.1, the SLR methodology involves defining keywords, searching databases, and applying filters to select relevant studies for review. The initial search was performed by searching the search string in the defined databases without applying any filters (i.e., the words in the search string could appear in "Full Text" – or "All Fields", depending on the nomenclature defined in the database). Following that, other filters were applied to further refine the results. Table 2.3 describes all the filters applied.

Table 2.2. Conducting the Review Stages

Stage	**Activities**
1. Identification of the Research	a. Define KWs based on Title and RQs. b. Define Search String (SS) based on KWs.
2. Study Selection	**Database Search** a. Access the chosen scientific databases. b. Conduct the search with the SS. c. Apply filters. **Snowball Search** a. Select relevant studies from references of the selected studies.
3. Study Quality Assessment	a. Interpret findings carefully.
4. Data extraction and monitoring progress	a. Capture relevant information from studies.
5. Data synthesis	a. Summarize the results of included studies.

[1]https://ieeexplore.ieee.org/
[2]https://scholar.google.com/
[3]https://www.scopus.com/
[4]https://www.webofscience.com/

Table 2.3. Filters applied to the search

Filter	Description
F0	Full Text, All Fields
F1	SK1.1
F2	Only in English
F3	Publication Year $\geq$ 2015
F4	Only Journal Articles or Conference Papers
F5	Subject Area: Comp. Science, Engineering, Business **(applied only in Scopus database)**
F6	Manual Selection
F7	Remove Duplicates

2.1.2. Conducting the Review

This section describes the second phase of the SLR methodology, detailing the conduction of the review based on the review protocol mentioned above and the analysis of the extracted data.

2.1.2.1. *Identification of the Review*

According to the research question defined, these were the Keyword (KW) used and combined to form the SS. Besides the KW defined in Table 2.4, synonyms for each word were used to amplify the search and gather more results.

Table 2.4. Definition of Keywords and Search String

Keywords	Topic Model; Hierarchical Topic Model; Knowledge Representation; Knowledge Extraction; Education
SS1	"(("topic model" OR "topic modeling" OR "hierarchical topic model" OR "hierarchical topic modeling") AND("knowledge representation" OR "knowledge extraction" OR "knowledge mining" OR "knowledge graph" OR "knowledge graphs") AND ("education" OR "educational content"))"
SS1.1	"(ABS("topic model" OR "topic modeling" OR "hierarchical topic model" OR "hierarchical topic modeling") AND ABS("knowledge representation" OR "knowledge extraction" OR "knowledge mining" OR "knowledge graph" OR "knowledge graphs") AND ALL("education" OR "educational content"))"

2.1.2.2. *Study Selection*

The first and most important stage of the review process was the database search. A total of 3101 documents were initially retrieved by applying F0 (searching in "Full Text" or "All Fields") across four major digital libraries: IEEE Xplore (434), ACM Digital Library (915), Scopus (1739), and Web of Science (13).

To narrow the scope, the F1 filter applied a predefined keyword string (SS1.1), reducing the dataset to 79 studies. Subsequently, F2 was applied to include only publications

written in English, resulting in 75 studies. The third filtering step (F3) further excluded studies published prior to 2015, maintaining a total of 75. Next, F4 limited the selection to Journal Articles or Conference Papers, which again retained 75 studies. Then, F5, applied exclusively to the Scopus database, filtered for documents in the subject areas of Computer Science, Engineering, and Business, reducing the dataset to 69 studies.

In the final phase of selection, filters F6 (Manual Selection) and F7 (Duplicate Removal) were applied simultaneously using Zotero. In this step, duplicates were merged automatically, and irrelevant documents were manually excluded based on abstract reading. This process reduced the corpus from 64 to 40 studies, and 7 duplicate documents were excluded during manual screening, resulting in a total of 33 studies.

This step-by-step approach ensured transparency, reproducibility, and methodological rigor, in line with the SLR guidelines proposed by the author of [4].

Table 2.5. Number of results after the application of each filter described in Table 2.3

DB/Filters	**F0**	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**
IEEE	434	12	12	12	11	11	9	8
ACM	945	9	9	9	9	9	8	6
Scopus	1739	52	48	48	43	38	20	17
Web of Science	13	6	6	6	6	6	3	2
Total	3101	79	75	75	69	64	40	**33**

In addition to the database search, a Snowballing Search technique is being conducted in parallel throughout the development of this research. This strategy is meant to identify potentially relevant studies that may not have been captured through the initial search queries. As a result, the final number of included articles may be revised in the end. Table x and x details the quantitative number of studies retrieved from the SLR and the current count of documents from the snowballing technique.

Table 2.6. Total count of documents — with Snowballing Results

Database Search	33
Snowballing Search	9
Total	**42**

2.1.2.3. *Data Extraction and Primary Analysis*

The publications included in the analysis (Figure 2.1) were mainly Journal Articles, comprising 61% of the total, while Conference Papers accounted for the remaining 39%. This distribution indicates a strong level of academic development in the field, as Journal Articles generally require more strict peer-review processes than Conference Papers [8].

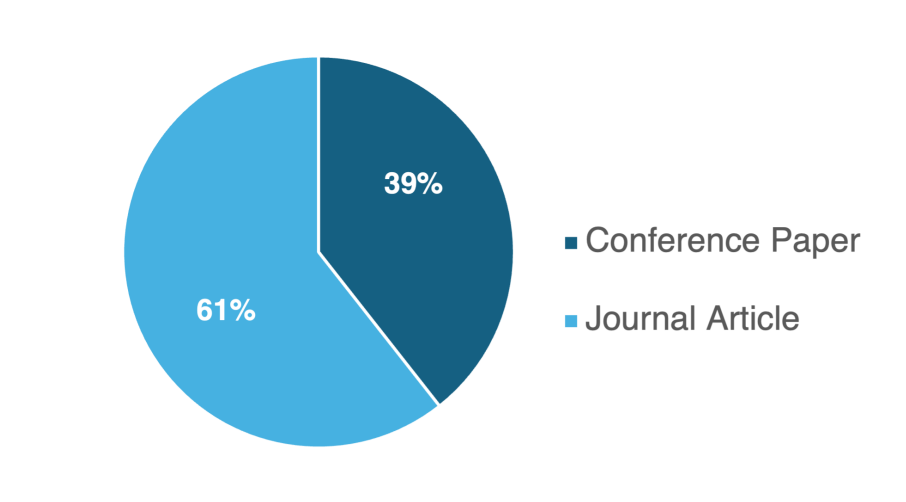

FIGURE 2.1. Pie Chart of Selected Studies Based on Study Type

Figure 2.2 shows the distribution of the articles found per year. Most of the studies were published between 2020 and 2023, with 2023 being the year with the highest number of publications (9 studies). There has been a noticeable increase in research activity since 2019, reflecting a growing interest in this area over the past five years. In contrast, earlier years, especially before 2017, contributed very few studies to the dataset, underlining the recent relevance of the field.

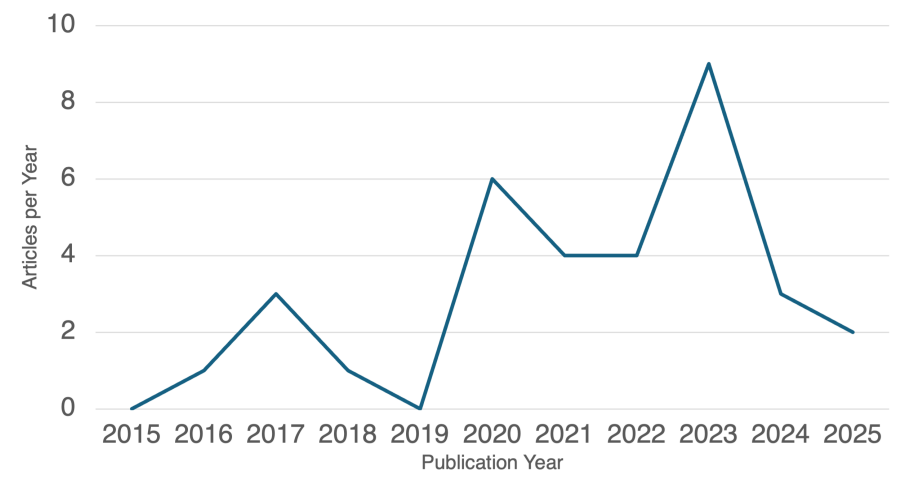

FIGURE 2.2. Line Chart of Selected Studies Based on Study's Publication Year

Beyond the quantitative distribution of studies, it is essential to undertake a qualitative analysis that clarifies the substantive focus of the X articles under review. To this end, two concept centric tables were constructed. The first table categorizes the literature according to application domains, with education emerging as the most prominent area, followed by domains such as social media, healthcare, legal, and others. This table further demonstrates how topic modeling and knowledge representation techniques have been adapted to address the unique challenges inherent to each field. The second

table organizes the studies by thematic focus—such as Topic Modeling (TM), Knowledge Graph (KG), and Hybrid Approaches, and others—and specific technical implementations, including Latent Dirichlet Allocation (LDA), Ontologies, Embeddings, BERT, and Cosine Similarities.

Overall, these concept centric tables offer a macro-level overview of the main topics of this study, separated by Domain and Field, helping to identify patterns of methodological convergence, recurring technical decisions, and domain-specific adaptations across the selected literature.

Table 2.7. Analysis of Domain-Specific Applications and Key Features

Domain	**Features**	**References**
Education	Course Recommendations; Educational Knowledge Graphs; Instructional Topic Extraction; E-learning Evaluation; Topic-based Document Retrieval; Domain-specific QA Resources	[9]–[14]
Social Media/News	Hate-Speech and Inequality Detection; Sentiment Knowledge Graphs; Short/Noisy Text Streams; Entity Linking in News & Social Media	[15]–[20]
Business	Knowledge Graph-Based Recommendations; Business Trajectory Analysis; Topic Mining in E-Commerce Corpora	[21]–[24]
Legal	Legal Knowledge Graphs for CFR/FARS/SLAs; Heterogeneous KG Generation for Criminology	[25], [26]
Healthcare	Lung Cancer Knowledge Classification; Knowledge Acquisition Across Medical Subdomains	[17], [27], [28]
Environmental Sciences	Machine Learning for Climate Bibliometrics and Topic Trends	[29]
Literature	Analysis of Chinese Literature: Topics and Relationships	[30], [31]
Internet of Things (IoT)	IoT Service Matching	[32]
Argument Mining & Decision Support	Topic-Guided KG Construction for Arguments and Decision Support	[33]
Multidisciplinary	Text Clustering; Cross-Lingual Alignment; Hierarchical Topic Modeling; KGE-LDA; Entity Retrieval; Cross-Discipline Trend Analysis	[31], [34]–[42]

As shown in Table 2.7, most advancements are still concentrated in foundational methods and generic applications, as evidenced by the high volume of work in Multidisciplinary research (e.g., [31], [38], [39], [42]). This focus extends to other high-activity domains such as Social Media/News (e.g., [15], [17], [19]), Business (e.g., [21], [22]), and Healthcare (e.g.,

[27], [28]). In contrast, Education (e.g., [9], [10], [13]), while an active area, remains a relatively less-explored field for these advanced integrations, despite its high-impact potential. Furthermore, persistent challenges identified in the literature are reflected across domains, including handling short and noisy text streams [17], managing semantic heterogeneity [26], [28], and the difficulty of adapting models to multidisciplinary materials [41].

Overall, the reviewed studies provide a solid foundation for this research. They demonstrate methodological advances and interdisciplinary potential by combining Topic Modeling (e.g., [10], [21], [31]) and Knowledge Representation (e.g., [9], [16], [22], [25]). More importantly, specific work on integrating these two fields [33], [39], highlights existing precedent and validates the approach. These findings justify the need for a methodology that not only leverages state-of-the-art HTM [31], [38], but also explores its integration with KR to support the specific needs of Educational Applications [10], [14].

Table 2.8. Strategic Overview of Techniques by Research Domain

Research Domain	**Techniques**	**References**
Topic Modeling	Classic Topic Modeling (LDA)	[9]–[13], [17]–[19], [22], [24]–[27], [29]–[35], [37], [39]–[42]
	Word Embeddings	[13], [17], [18], [24], [33], [35], [39]
	Clustering Algorithms	[24], [31], [32], [34], [37], [41]
Knowledge Graphs	Ontologies	[12], [34]
Hybrid Approaches	Topic Modeling + KG Embeddings	[11], [22], [36], [39], [42]
	Ontologies + Semantic Web	[34], [41]
	GNN + Topic Modeling + KGs	[18], [27]
	Classic LDA + Knowledge Graphs	[25], [27], [34], [35], [37]
	BERT-based (Transformers/Clustering)	[11], [14], [23], [35], [41]
	TF-IDF	[10], [12], [28], [29], [33], [35]
Entity Alignment	KG Embeddings + Biterm TM	[36], [42]
Recommender Systems	Cosine Similarity	[9]
	Relation-Aware GNN (RAGNN)	[21]
	LDA	[9], [21]
Sentiment Analysis	Sentimental KG	[35]
	Topic Modeling	[15]
	Hybrid (TM + Sentimental KG)	[11], [17]

As outlined in Table 2.8, the literature is underpinned by two core methodologies: Topic Modeling (TM) and Knowledge Graph (KG). In the realm of TM, Latent Dirichlet Allocation (LDA) serves as a cornerstone technique, collecting frequent citation in several investigations [9]–[13], [17]–[19], [22], [24]–[27], [29]–[35], [37], [39]–[42]. This traditional method is often augmented by Word Embeddings [13], [17], [18], [24], [33], [35], [39] and Clustering Algorithms [24], [31], [32], [34], [37], [41] to support semantic comprehension. In parallel to this, the domain of KG highlights considerable research on Ontologies [12], [13], [16], [26], [31], [34], [37], [41].

A notable and a significant trend observed in the literature is the integration of these fields into Hybrid Approaches. This integration is characterized by the fusion of Topic Modeling (TM) with Knowledge Graph Embeddings (KGE) [11], [22], [36], [39], [42] and the combination of Ontologies with Semantic Web technologies [34], [41]. More complex architectures have also surfaced, such as pairing **GNN!** (**GNN!**) with TM and KG [18], [27]. However, traditional methods remain relevant to these systems, with LDA frequently integrated with KG as shown by the number of studies found in the SLR [25], [27], [34], [35], [37]. Additionally, the adoption of Transformer-based models combined with clustering [11], [14], [23], [35], [41] suggests a shift towards contextual representations, even as foundational techniques like Term Frequency-Inverse Document Frequency (TF-IDF) continue to be implemented for baseline comparison [10], [12], [28], [29], [33], [35].

Ultimately, these methodologies can also be adapted for specific downstream tasks. The literature demonstrates their application in Entity Alignment [36], [42] and Recommender Systems [9], [21], as well as Sentiment Analysis [11], [15], [17], [35], where approaches range from standalone topic modeling to hybrid graph integrations.

The following sections provide the theoretical groundwork for this study. The review covers TM (Section 2.2) and KR (Section 2.3), followed by an analysis of their integration (Section 2.4). Finally, specific attention is given to the challenges of modeling educational content (Section 2.5), which constitutes the primary domain of this work.

2.2. Topic Modeling (TM)

TM refers to a combination of unsupervised Machine Learning (ML) techniques used to uncover latent thematic structures within large collections of textual data, while ensuring the correct statistical correlations. It is based on the idea that documents are composed of multiple hidden topics, where each topic is a probability distribution over words, and each document is a mixture of these topics [43], [44]. Unlike traditional vector space models such as TF-IDF or Bag of Words (BoW), which often result in superficial and semantically shallow representations [34], [45], TM facilitates deeper semantic understanding and more meaningful clustering, classifications, and summarization of textual data.

One of the foundational statistical models in this field is Probabilistic Latent Semantic Analysis (PLSA), introduced by Hofmann [46]. PLSA models the co-occurrence of words and documents through a latent variable framework, in which each word in a document is assumed to be generated from a latent topic variable. While PLSA marked a significant improvement over classical methods like Latent Semantic Analysis (LSA) [47], it lacks a fully generative structure for documents, making it prone to overfitting and incapable of modeling unseen documents without re-training the entire model.

Such limitations were addressed by the development of LDA, proposed by Blei, Ng, and Jordan [44]. LDA employs Dirichlet priors [48] to probabilistically model the distribution of topics within documents and words within topics, enabling a generative approach that effectively captures latent structures and generalizes to previously unseen data. More specifically, in LDA, the generative process involves sampling a distribution over topics

for each document and, for each word, selecting a topic, and then sampling a word from the corresponding topic distribution. Hence, this structure has made LDA one of the most widely adopted models in various domains such as Media [17], [19], Internet of Things (IoT) [32], Health [27], [28], [37], Legal [25], [29], Education [9]–[13], and Multidisciplinary [40], [41].

Despite its reliability, LDA and other similar models assume a flat topic structure, which limits their capability when dealing with hierarchical collections. Thus, Griffiths et al. [49] proposed a HTM, named hierarchical Latent Dirichlet Allocation (hLDA), where topics are organized into a tree-like structure that captures specific semantic relationships among them. The hLDA model combines LDA with a nonparametric prior, known as the Chinese Restaurant Process (CRP) [50], which was then extended, and referred to as nested CRP. This prior enables the discovery of topic hierarchies without predefining the number of topics nor the depth of the tree, and in hLDA, each document is modeled as a path through an infinite tree, sampling words from topic distributions associated with nodes along that path. Moreover, a Gibbs sampling algorithm is implemented for inference, allowing the model to simultaneously explore both the topic content and hierarchical structure. On the other hand, these models often constrained each document to a single path, restricting their capacity to model documents that span multiple thematic areas.

On top of that, Shin and Moon [38] identify another limitation of hLDA hLDA [49]: its inability to integrate prior-knowledge or domain-specific knowledge during the topic inference process, often leading to abstract topic structures that lack semantic coherence. In contrast, Shin and Moon have sought to address this limitation by introducing Guided Hierarchical Topic Model (GHTM), which offers a more robust approach by explicitly merging HTM with structured knowledge representation. GHTM is the first model to apply the Dirichlet Forest prior, an advancement of the traditional Dirichlet prior [48] that incorporates semantic constraints through a forest-like structure of topic paths. This approach allows documents to be associated with multiple topic hierarchies and incorporates prior knowledge through *must-link* and *cannot-link* clustering relationships, guiding the model in grouping semantically related words while effectively distinguishing unrelated ones.

By incorporating these constraints, GHTM represents sort of a hybrid approach of TM and KR (explained in more detail in Section 2.3). This model aligns with broader trends in semantic modeling, where structured knowledge, often represented through Ontologies or KGs, is employed to enhance interpretability and relevance of machine-generated topics. The guided facilitates the development of topic hierarchies that not only reflect statistical patterns in the data but also adhere to domain ontologies or expert knowledge, resulting in topics that are both semantically meaningful and pedagogically relevant. Overall, the empirical evaluations suggest that GHTM outperforms traditional HTMs in key metrics such as F-measure, perplexity, and KL-divergence-based separability.

The progression towards more expressive TMs has also led to hybrid frameworks that incorporate external semantic knowledge. For instance, Fatimi et al. [34] proposed a framework for semantic clustering where unstructured text is transformed into RDF triples and enhanced using background knowledge to better capture meaning, supporting better topic interpretation and domain-specific reasoning. Likewise, Linh et al. [18] introduced a Graph Convolutional Topic Model (GCTM) that enhances topic inference for short and noisy text streams-an environment where traditional models often underperform. GCTM incorporates external word knowledge graphs, such as WordNet or embeddings like Word2Vec, and applies graph convolutional layers to strengthen co-occurrence patterns by uncovering deeper semantic connections. This results in more accurate and coherent topic structures, even under concept drift or evolving data.

Moreover, recent advances in topic modeling have embraced neural and transformer-based methods to improve topic coherence and adaptability. As an example, Ennajari et al. [35] proposed a spherical topic model that integrates knowledge graphs and word embeddings, enhancing semantic structure in the latent space. Another work [14] applied few-shot relation learning using transformer architectures to extract meaningful relations from limited data, improving knowledge-aware topic extraction. Additionally, Arslan and Cruz [23] used BERTopic with controlled vocabularies to model business trajectories, enabling semantically enriched, domain-independent topic representations. Hence, these approaches reflect a trend towards hybrid models that combine structured knowledge with contextual embeddings.

In summary, topic modeling has undergone a considerable improvement—from flat probabilistic models like PLSA and LDA, to hierarchical frameworks such as hLDA and GHTM, and more recently, to hybrid and neural architectures that incorporate semantic knowledge and deep contextual embeddings. These advancements not only overcome previous limitations related to model inflexibility and topic interpretability, but also improve the potential of TMs strategies to function effectively across diverse and complex domains. As a result, modern TMs techniques are currently capable of capturing rich semantic structures, adapting to noisy or evolving corpora, and integrating domain-specific knowledge, thus reinforcing their relevance and applicability in information retrieval, decision-support systems, and knowledge representation.

2.3. Knowledge Representation

According to Nguyen et al. [12], KR refers to the structured modeling of information using logical components that make the semantics of a specific domain or topic machine-interpretable. This is increasingly important for improving the effectiveness of document retrieval systems and supporting different semantic applications. By encoding the meaning, content, and application code of textual data separately, KR allows machines to comprehend textual data, exchange understandings, and make inferences.

Among the most common knowledge structures, Ontologies, and KGs are frequently used. Ontologies provide the formal schema and logical underpinnings, while KGs instantiate these schemas with real-world entities and relationships. Both of them offer a robust practice for organizing, structuring, and analyzing large amounts of data that have complex relationships [15], [27].

2.3.1. Ontologies

An ontology is defined as a shared vocabulary and common information within a specific knowledge domain, encompassing definitions of fundamental concepts that can be interpreted by computers for understanding within that specific field. Ontologies serve as the blueprint for constructing KGs, while KGs capture the actual data, ontologies define the rules and vocabulary that structure this data [12], [34], [51].

Languages such as the Web Ontology Language (OWL) [52] are commonly used to define ontologies, offering a formal semantics grounded in first-order logic and model theory, which allows automated reasoning and inference. For example, in the educational domain, entities like *Course*, *Student*, and *Assessment* may be defined, along with relationships such as *enrolledIn* or *hasGrade*. These formal representations enable semantic interoperability, allowing heterogeneous systems to integrate, share, and reason over domain knowledge effectively.

2.3.2. Knowledge Graphs

A Knowledge Graph (KG) is a network representation of real-world entities (such as objects, events, and people) and the relationships encoded between them [53]. KGs are typically composed of three main components:

- **Nodes:** Represents entities or concepts (e.g, people, events, objects)
- **Edges:** Represents semantic relationships between entities
- **Labels:** Describes Nodes and Edges, enriches the graph with semantic context and enables interpretability by both humans and machines

Additionally, they are often structured as directed heterogeneous graphs where nodes and edges indicate entities and semantic linkages. Therefore, a KG can be described as a collection of triples (h, r, t), where 'h' is the head entity, 't' is the tail entity, and 'r' is the relationship linking them [35]. They organize data from multiple sources and capture information about entities of interest, forging connections between them.

In summary, Ontologies and KGs together form the backbone of modern knowledge representation strategies that enable intelligent systems to effectively utilize and process knowledge information [28]. Ontologies provide the formal, machine-interpretable definitions and semantic structure required to model domain knowledge with precision and consistency. Building upon this foundation, KGs enable the practical organization of real-world data through richly interconnected entities and relationships [34]. Thus, when combined, they enable reasoning, data integration, and semantic interoperability across

various sources, enabling intelligent systems to comprehend, relate, and act upon complex information in a meaningful and scalable manner. This can be used to enhance various tasks such as question answering [13], recommendation systems [21], and information retrieval[9].

2.4. Topic Modeling and Knowledge Representation

Following the described background of TM and KR, it is possible to assume the these two fields are profoundly interconnected that, when combined, offer a mroe sophisticated and interpretable understanding and organization of information, moving beyond simple text analysis, to capturing, for instance, rich semantic meaning and relationships [34].

Moreover, this integration enhances the interpretability and semantic coherence of topics discovered by topic models, which can otherwise be unclear or difficult to interpret due to their lack of semantic context or prior knowledge [12]. By incorporating external knowledge from KGs or ontologies, TM can infer more accurate, coherent, and semantically meaningful topics [35], [39]. For instance, KGE, which map entities and relationships into low-dimensional vectors while preserving the graph's structure, can guide the inference process of semantically coherent topics. Moreover, hybrid models like BERTopic [23] leverage contextualized language representations to consider underlying semantic context and create denser clusters of interpretable topics.

In conclusion, these works showcase that topic modeling has the potential to be helpful in the construction and enrichment of KGs and ontologies from unstructured text by identifying relevant thematic concepts, key terms, and summarizing content to build structured representations. A few examples include LDA for classifying medical articles into topics for lung cancer knowledge classification [27], or for building KGs to analyze narrative structures in literary works [30].

2.5. Challenges in Modeling Educational Content

The modeling of educational content—particularly for purposes such as knowledge representation, recommendation, or test construction—presents a complex array of challenges that span semantic ambiguity, structural inconsistency, and the lack of ground-truth evaluation [1]–[3].

A recurrent challenge in the context of education is the absence of a structured framework oriented to specific domains. Educational resources, including lecture slides and exam questions, frequently exhibit inconsistent formatting and a lack of semantic unity. This inconsistency poses significant challenges for the implementation of conventional topic modeling techniques. In response to these issues, Xue et al. [10] introduced a dynamic educational knowledge graph model that utilizes information entropy to tackle the complexities of knowledge evolution and concept drift within learning environments. Although this model enhanced the adaptability of content representation, it still faced

limitations related to the initial coverage of the knowledge base and its reliance on high-quality labeled data. Thus, this indicates a need for further investigation into unsupervised methodologies to overcome these constraints.

Another challenge concerns content sparsity and incoherence in short texts, such as *quiz items* or *slide bullets*. Shin and Moon addressed this by developing Guided Hierarchical Topic Model (GHTM)—previously described in Section 2.2, which integrates domain knowledge through Dirichlet Forest priors to structure topics hierarchically. Their approach demonstrated improved clustering and interpretability; however, it also relied heavily on the quality and specificity of the seed words, making its adaptability to new or interdisciplinary subjects limited. In conclusion, they acknowledged that future research should refine seed word selection strategies and develop automated methods for domain knowledge extraction

Additionally, semantic gap and heterogeneity across learning objects presents a barrier to uniform representation. Yi et al. [11] incorporated sentiment analysis and KGs to enhance e-learning evaluation, revealing how emotionally annotated feedback could contextualize student engagement. Yet, their system struggled to integrate heterogeneous learning resources such as slides, forums, and quizzes due to the lack of a unified semantic schema. s In more specialized domains like science education, Zhou et al. [13] developed a Question Answering (QA) system for chemistry education that mapped questions to structured knowledge graphs. This approach improved precision in response generation, but faced challenges in scalability and adaptability across diverse subfields within chemistry, highlighting the need for more modular and cross-domain compatible models.

2.6. Summary

In summary, the literature review suggests a research landscape that has evolved significantly over the past years. TM techniques have progressed from classic probabilistic models (PLSA, LDA) to hierarchical structures (hLDA, GHTM) and, more recently, to hybrid and neural-based frameworks—integrating semantic knowledge through ontologies, knowledge graphs, and embeddings. This reflects a continuous effort to overcome the limitations of classic unsupervised probabilistic models.

In parallel, KR has developed as a field dedicated to structuring and formalizing domain knowledge in a machine-interpretable way. Ontologies and KGs emerged as the core mechanisms supporting this objective: ontologies provide formal semantics and conceptual frameworks, while KGs instantiate these abstractions into interconnected networks of entities and relations.

The combination of TM and KR illustrates the value of hybrid strategies in connecting purely data-driven statistical techniques with well-defined, domain-oriented semantic frameworks. By merging the probabilistic reasoning capabilities of TMs with the clear relational frameworks provided by ontologies and KGs, these approaches ease the identification of patterns while safeguarding the integrity of concepts. Additionally, this intersection improves the clarity, consistency, and compatibility of topics with domain

knowledge, overcoming a primary drawback of conventional probabilistic methods. Additionally, imminent research utilizes KGE and contextual embeddings to inject semantic priors into the modeling process, resulting in topics that not only reflect statistical patterns but also adhere to conceptual hierarchies and well-defined relations. Therefore, this integration supports automatic construction, enrichment, and reasoning over structured knowledge bases, thereby enhancing explainability and adaptability across diverse and evolving domains.

CHAPTER 3

Methodology

As introduced in Chapter 1, we plan to follow a CRISP-DM methodology to guide or process. Proposed by Wirth and Hipp [6], the CRISP-DM is a framework that structures data mining projects into phases: (1) business understanding, (2) data understanding, (3) data preparation, (4) modeling, (5) evaluation, and (6) deployment. It emphasizes iterative development and feedback loops between phases. By applying CRISP-DM to our solution, the objective is to document each step thoroughly in order to inform future iterations and incrementally build tasks that address our specific context, such as dealing with multimodal content and integrating TM outputs with KR structures.

We intend to undertake fundamental preprocessing steps to transform lecture slides into machine-readable text, identify potential data quality issues, and prepare initial datasets. Following that, we plan to explore and test HTM techniques preliminary, without committing to a definitive modeling strategy at this stage. Additionally, we aim to qualitatively assess the meaningfulness of the extracted topics and investigate whether the integration of knowledge representation methods, such as simple graphs or ontologies, could enhance semantic coherence.

Finally, as this proposal is in its early stages, we acknowledge that the methodology may undergo modifications. Further iterations of data understanding, modeling, and evaluation will be necessary, in keeping with the iterative and flexible nature of CRISP-DM, to refine our approach as new insights are gained.

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